PHYSICAL CHEMISTRY CUMULATIVE EXAM

October 3, 1998

1. (a) Consider a photochemical reaction in which one molecule of A will react for every photon of light energy absorbed:

$A + h\nu \rightarrow$ Products

Assume that five photons are being absorbed each second.
If the initial concentration $C_A = 1M$, what will be the rate of the reaction?
If the initial concentration $C_A = 2M$, what will be the rate of the reaction?
What is the order of the reaction?

(b) Using the equation for the half-life of reaction:

$$\log(t_{½}) = \log[(2^{n-1} - 1)/(n - 1)k] - (n - 1)\log(C_i)$$

determine the order of the reaction for which the half-life is halved as the initial concentration of reactant is doubled. C_i is the initial concentration.

2. On planet X, chemistry is different from that on earth. A chemist there finds kinetic data for a reaction that suggests that a 20 degree rise in temperature increases the rate of a reaction by a factor of 3. He then writes in a text book on Chemical Kinetics, "For chemical reactions, a 20 degree rise in temperature increases the rate by a factor of 3".

Under what conditions would this statement be true? Would this statement apply to all reactions on planet X (explain your answer in detail)?